

PAPER_319: Compact HII SFR Gravitational Binding Phase Transition

t_cross = 67,730 yr | sSFR = $5 \times 10^{-4} \text{ yr}^{-1}$ (50× Lagoon) | m_factor(t_age) = 151

FIRST UQFF Compact HII Region SFR Runaway Gravitational Binding Phase Transition

Session: 91

Module: ORION_UQFF_MODULE.cpp (33rd C++ module)

System: Orion Nebula M42/NGC 1976 — compact HII region with active star formation

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Abstract

This paper derives the SFR-driven gravitational binding phase transition for the Orion Nebula. With SFR = 1 M_{sun}/yr acting on an initial mass M = 2000 M_{sun}, the specific star formation rate **sSFR = $5 \times 10^{-4} \text{ yr}^{-1}$ is 50× that of the Lagoon Nebula** ($1 \times 10^{-5} \text{ yr}^{-1}$, PAPER_305). The system is born wind-dominated (unbound), but as SFR continuously adds mass, the effective gravitational acceleration amplified by the SFR mass factor $m_factor(t) = 1 + \text{SFR} \times t_yr / M_sun_count$ crosses the growing wind ram pressure at **t_cross = 67,730 yr**, transitioning the nebula from unbound to gravitationally bound. By t_age = 300 kyr, $\text{binding_ratio} = g_SFR / a_wind = 2.654$. This is the FIRST UQFF compact HII SFR runaway gravitational binding phase transition.

System Parameters

Parameter	Value	Description
M	2000 M _{sun} = $3.978 \times 10^{33} \text{ kg}$	Initial mass
SFR	1 M _{sun} /yr = $6.303 \times 10^{22} \text{ kg/s}$	Star formation rate
sSFR	$5 \times 10^{-4} \text{ yr}^{-1}$	Specific SFR = SFR/M
t_age	$3 \times 10^5 \text{ yr}$	Nebula age
v_wind	$8 \times 10^3 \text{ m/s}$	HII ionization front
g_base	$1.907 \times 10^{-11} \text{ m/s}^2$	Base gravity (PAPER_317)
a_wind(t=0)	$5.424 \times 10^{-10} \text{ m/s}^2$	Initial wind acceleration

Key Equations

SFR Mass Factor

$$M_{\text{sf}}(t) = \frac{\text{SFR}_{\text{yr}} \times t_{\text{yr}}}{M_{\text{sun,count}}}, \quad m_{\text{factor}}(t) = 1 + M_{\text{sf}}(t)$$

SFR-Amplified Gravitational Acceleration

$$g_{\text{SFR}}(t) = g_{\text{base}} \times m_{\text{factor}}(t) = g_{\text{base}} \left(1 + \frac{\text{SFR}_{\text{yr}} \cdot t_{\text{yr}}}{M_{\text{sun,count}}} \right)$$

Wind Acceleration (time-evolving)

$$a_{\text{wind}}(t) = \frac{v_{\text{wind}}^2}{r} \left(1 + \frac{t}{t_{\text{age}}} \right)$$

Crossover Time t_{cross} (PAPER_319 Key Result)

Setting $g_{\text{SFR}}(t) = a_{\text{wind}}(t)$:

$$g_{\text{base}} (1 + \text{sSFR} \cdot t) = a_{\text{wind},0} \left(1 + \frac{t}{t_{\text{age,yr}}} \right)$$

$$t_{\text{cross}} = \frac{a_{\text{wind},0} - g_{\text{base}}}{g_{\text{base}} \cdot \text{sSFR} - a_{\text{wind},0}/t_{\text{age,yr}}} = \boxed{67,730 \text{ yr}}$$

Specific SFR

$$\text{sSFR} = \frac{\text{SFR}_{\text{yr}}}{M_{\text{sun,count}}} = \frac{1}{2000} = 5 \times 10^{-4} \text{ yr}^{-1} = 50 \times \text{Lagoon}$$

Gas Depletion Timescale

$$t_{\text{consume}} = \frac{M_{\text{sun,count}}}{\text{SFR}_{\text{yr}}} = \frac{2000}{1} = 2000 \text{ yr} \quad (\text{without OMC-1 replenishment})$$

Results Summary

Quantity	Value	Significance
sSFR	$5 \times 10^{-4} \text{ yr}^{-1}$	50× Lagoon Nebula
t_{cross}	67,730 yr	Unbound → bound transition
$m_{\text{factor}}(t_{\text{age}}=300 \text{ kyr})$	151	SFR amplification
$g_{\text{SFR}}(t_{\text{age}})$	$2.878 \times 10^{-9} \text{ m/s}^2$	$151 \times g_{\text{base}}$

Quantity	Value	Significance
a_wind(t_age)	$1.085 \times 10^{-9} \text{ m/s}^2$	
binding_ratio(t_age)	2.654	Gravitationally bound
m_factor(1 Myr)	501	
binding_ratio(1 Myr)	4.069	Increasingly bound
t_consume	2000 yr	Gas depletion (no replenishment)

Phase Transition Diagram

t=0	t_cross=67.7 kyr	t_age=300 kyr	t=1 Myr
UNBOUND	TRANSITION	BOUND 2.65×	BOUND 4.07×
a_wind > g_SFR	a_wind = g_SFR	g_SFR > a_wind	g_SFR >> a_wind
$\eta = 28.47 \text{ wind}$	— crossover —	$\eta_B = 2.654$	$\eta_B = 4.069$

Comparison: UQFF SFR Class

Module	SFR (M_{sun}/yr)	M (M_{sun})	sSFR (yr^{-1})	PAPER
Lagoon	0.1	10,000	1×10^{-5}	PAPER_305
Orion	1.0	2000	5×10^{-4}	PAPER_319
M16 (Eagle)	$\sim 0.01 \times$	1200	—	PAPER_284

Orion sSFR is $50 \times$ Lagoon — UQFF identifies this as the “ultra-compact HII” class with rapid gravitational binding crossover. Lagoon has $t_{\text{consume}} = 100 \text{ kyr}$; Orion has $t_{\text{consume}} = \mathbf{2000 \text{ yr}}$, the shortest gas depletion time in the UQFF series, sustained only by continuous OMC-1 giant molecular cloud inflow.

UQFF Physical Interpretation

The phase transition at $t_{\text{cross}} = 67,730 \text{ yr}$ is a structural boundary in Orion’s evolution:

- **Before t_{cross} :** The system is wind-dominated ($\eta_{\text{wind}} > 1$). UV photoionization and ram pressure drive a champagne flow outward; newly formed stars are subject to wind erosion.
- **After t_{cross} :** SFR has added sufficient mass that $g_{\text{SFR}} > a_{\text{wind}}$. The system becomes self-gravitating with respect to its own stellar formation. The cluster proceeds to form stars under its own gravity — consistent with the Orion OB1 association framework where the cluster is now 300 kyr old and gravitationally bound.

Within ORION_UQFF_MODULE.cpp, the SFR mass factor $m_factor(t)$ enters the base gravity term via $G*M*(1+M_sf(t))/r^2$, registered as WOLFRAM_TERM_ORION_BASE and WOLFRAM_TERM_ORION_SFR_BINDING.

WOLFRAM_TERM Registration

```
#define WOLFRAM_TERM_ORION_BASE(val)      (val)
// g_base = (G*M*(1+M_sf(t))/r^2)*(1+Hz*t)*(1-B/B_crit)*(1+f_TRZ)
// M_sf(t) = SFR_yr*t_yr / M_sun_count [PAPER_319 SFR amplification]

#define WOLFRAM_TERM_ORION_SFR_BINDING(val) (val)
// wind = v_wind^2/r * (1+t/t_age) [crosses SFR gravity at t_cross=67730 yr]
```

Series first: FIRST UQFF compact HII SFR runaway gravitational binding phase transition. Establishes sSFR as a new UQFF classification axis for HII regions: compact (Orion, $sSFR=5 \times 10^{-4} \text{ yr}^{-1}$) vs extended (Lagoon, $sSFR=1 \times 10^{-5} \text{ yr}^{-1}$).